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Towards Detecting LLMs Hallucination via Markov Chain-based Multi-agent Debate Framework

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***Abstract*—The advent of large language models has facilitated the development of natural language text generation. It also poses unprecedented challenges, with content hallucination emerging as a significant concern. Existing solutions often involve expensive and complex interventions during the training process. Moreover, some approaches emphasize problem disassembly while neglecting the crucial validation process, leading to performance degradation or limited applications. To overcome these limitations, we propose a Markov Chain-based multi-agent debate verification framework to enhance hallucination detection accuracy in concise claims. Our method integrates the fact-checking process, including claim detection, evidence retrieval, and multi-agent verification. In the verification stage, we deploy multiple agents through flexible Markov Chain-based debates to validate individual claims, ensuring meticulous verification outcomes. Experimental results across three generative tasks demonstrate that our approach achieves significant improvements over baselines.**

***Index Terms*—Hallucination Detection, Markov Chain, Multi-agent.**

I. Introduction

The continuous evolution of large language models (LLMs) has significantly expanded language processing capabilities across diverse domains [1], [2]. However, this progress introduces challenges, such as the substantial cost associated with updating model parameters and inherent deficiencies in reasoning [3]–[5]. This has led to the generation of inaccurate content, known as hallucination, particularly concerning potent yet opaque models like ChatGPT and GPT-4 [6].

Hallucination detection has become a focal point in addressing these challenges. Existing methods often necessitate costly and intricate interventions during the training process [7]–[9], rendering them unsuitable for large language models with agnostic parameters and these methods often incur considerable costs. Consequently, researchers have explored post-processing approaches [10]–[14] involving hallucination detection or correction post-content generation. Notably, these methods typically focus on problem decomposition and evidence retrieval, emphasizing simple prompting during individual verification. We posit that the verification accuracy is pivotal compared to problem decomposition in LLMs.

To address these challenges, we present a fact-checking process to enhance the accuracy of hallucination detection. As shown in Figure 1, which involves three stages: claim detection, evidence retrieval, and multi-agent verification. In claim detection, our approach involves the extraction of claims from extensive responses by prompting ChatGPT, decomposing the intricate problem into smaller components. Evidence retrieval involves generating queries based on claims for retrieval. Subsequently, we retrieve the corresponding evidence based on these generated queries. In the multi-agent verification stage, we innovatively propose a Markov Chain-based multi-agent debate verification framework, which leverages the robust capabilities of multi-agent systems to simulate human behavior. This approach involves deploying diverse agents in Markov Chain debates to verify individual claims, thus providing a nuanced and flexible validation process. Following the verification of each claim using our method, the collective judgment of all claims contributes to the detection of hallucinations in the original response.

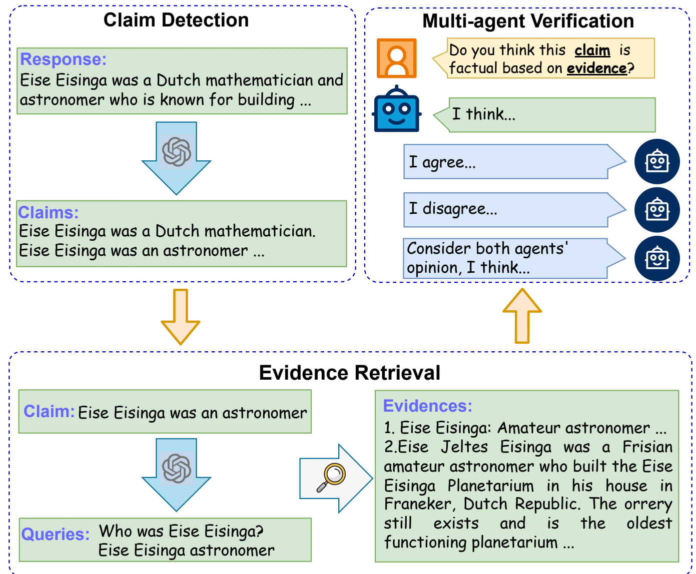

Fig. 1. Overview of the fact-checking process, which includes three distinct stages: **Claim Detection**, **Evidence Retrieval**, and **Multi-agent Verification**.

We conduct extensive experiments across three generative tasks demonstrating the effectiveness of our approach. Verification outcomes are meticulously analyzed and compared against existing methods to ascertain the superiority of our approach. In summary, our contributions can be summarized as follows:

- We propose a versatile hallucination detection process applicable to multiple generation tasks for improving verification accuracy.
- We introduce a Markov Chain-based multi-agent debate verification framework that simulates human discussion.
- Experiments conducted on three generative tasks show that our proposed framework outperforms baselines.

II. Related Work

A. Hallucination Detection

Detecting hallucinations in LLMs is crucial for ensuring their reliability. Numerous studies have employed fact-checking methodologies

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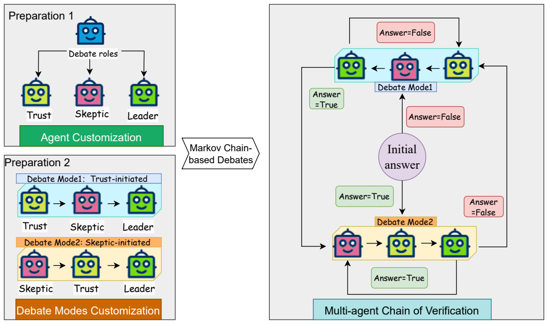

Fig. 2. Overview of the proposed multi-agent debate verification framework for hallucination detection.

to detect hallucinations. This was typically accomplished through three steps [15]: Claim Detection, Evidence Retrieval, and Verdict Prediction. With the advent of large language models, some works [10], [16] tackled the task of hallucination detection by prompting the large language models directly. In addition to task-specific approaches, there are hallucination detection methods specifically designed for LLMs. For example, some methods assess hallucination detection by examining the consistency of sampled examples [17], [18]. Our work is fundamentally based on the fact-check framework. We transfer the Verdict Prediction stage to the Multi-agent Verification to improve the precision of validation.

B. Multi-agent in LLMs

In recent years, there has been a significant increase in the size of models and the amount of training data used, resulting in the exceptional performance of large language models (LLMs) across various tasks. As a result, researchers have explored the use of LLMs as agents to simulate human behavior, leading to the development of influential projects such as Generative Agents [19], Ghost in the Minecraft [20], GPT-Bargaining [21] and Werewolf game [22]. There are also some efforts involve multiple agents engaging in debates to improve the reasoning capabilities [23]–[25] or address issues related to hallucinations [24], [26]. However, existing methods for hallucination detection and mitigation of LLMs solely rely on natural language interactions between agents, which may pose concerns regarding the self-correction approach [27]. Therefore, the objective of our work is to facilitate flexible discussions among multiple agents based on existing facts, aiming to detect and mitigate hallucinations in the generated content of language models.

III. Method

In this section, we introduce our method for detecting hallucinations in model-generated content. Our modified fact-checking process includes three stages: *Claim Detection, Evidence Retrieval, and Multi-agent Verification*. Verification errors often undermine overall effectiveness, despite accurate claims and evidence.

To address this, we propose a multi-agent debate verification framework, illustrated in Figure 2, using a Markov chain-based debate process to enhance verification accuracy. The following sections detail each stage, focusing on our innovative third stage.

A. Claim Detection

In the stage of claim detection, we employ the methodology utilized in Factool [12], leveraging large language models such as ChatGPT. Harnessing the robust instruction-following capabilities of LLMs empowers us to address the challenge of dissecting intricate responses. Nevertheless, detecting the hallucinations in statements lacking adequate information is futile and could impede overall judgment. Moreover, specific tasks may demand the concatenation of the model's responses with particular input information to formulate an informative claim, necessitating supplementary processing. Detailed explanations of these processing methods are provided in the experimental implementation section §IV-A2.

B. Evidence Retrieval

Upon extraction of claims, a retrieval methodology is employed to ascertain corresponding evidence. Drawing inspiration from Factool's [12] strategy in Knowledge Base Question Answering (KBQA) tasks, we prompt ChatGPT to formulate two queries, subsequently leveraging these queries to retrieve evidence. In instances where pertinent knowledge is absent, we employ the Google API to retrieve data from the internet. Conversely, when dealing with data accompanied by provided knowledge, we either consider the length of the knowledge as direct evidence or encode it for local retrieval.

C. Multi-agent Verification

We propose a Markov Chain-based multi-agent debate verification framework. Our investigations reveal significant potential in employing multi-agent systems to emulate human behavior [19], [20], particularly in the domain of fact-checking claims grounded in evidence. The key point in our method lies in the definition of **states** and the **transition** mechanisms.

1) Agents: To define the states, we must clarify the roles of the three agents: *Trust, Skeptic,* and *Leader*. These agents analyze perspectives based on claims and evidence from previous stages, either agreeing or disagreeing, and offering their own assessments. The **Trust** agent generally accepts the preceding agent's views, enhancing their credibility. The **Skeptic** agent challenges these views, seeking inconsistencies. The **Leader** agent synthesizes the perspectives of the Trust and Skeptic agents, evaluating both rational and irrational aspects to form its own viewpoint. Different personas for these agents are implemented through various prompts.

2) States: According to the definition of the Markov chain, we require an initial state to initiate our verification chain. Each agent analyzes preceding perspectives, starting from an initial state, denoted as S_0, where the initial agent provides the primary answer to initiate the verification chain.

We define two ordinary states, each involving three agents, representing distinct discussion modes. The first, labeled as S_1, begins with the Trust agent, followed by Skeptic and Leader, aiming to establish credibility before introducing skepticism. The second state, S_2, initiated by the Skeptic agent, follows the sequence Skeptic-Trust-Leader, emphasizing questioning credibility before further analysis. The chain alternates between these modes to reach a conclusive judgment.

To avoid indefinite extension, a termination state is crucial. Similar to human debates ending with aligned opinions, our termination condition activates if all three agents reach a consensus within a state. When the Skeptic agent finds no points of contention, and the Leader agrees without objections, the debate concludes. We also impose a maximum limit on verification rounds to control chain length.

3) Transition: Transitioning between states is a critical aspect of our methodology, following the definition of states. The primary criterion guiding these transitions in our approach is the verification result of a claim by the preceding state. This methodology aligns with human intuition, acknowledging the potential for diverse perspectives in debating a given matter.

Our transition probabilities are as follows:

$$Pr\left(S_2|R=True\right)=1 \tag{1}$$

$$Pr\left(S_1|R=False\right)=1 \tag{2}$$

R denotes the judgment from the preceding state. Our transition method operates as follows: if the previous state confirms the claim as factual, we transition to S_2 to rigorously analyze and question it, seeking loopholes. If none are found, the Trust agent accepts the answer, leading to chain convergence. Conversely, when the preceding state categorizes the claim as non-factual, we transition to S_1. In essence, we initially reinforce the credibility of this judgment, confirming the validity of skepticism. By enhancing the credibility of this opinion, if subsequent skepticism from the Skeptic agent is challenging, we can reasonably conclude the accuracy of this judgment, leading to the convergence of the chain.

Therefore, our process begins with an initial answer from S_0, followed by transitions to S_1 or S_2 based on this answer. Subsequent transitions depend on the preceding state's judgment, continuing until consensus within a state is reached, yielding the final verification result.

IV. Experiments

We conducted experiments encompassing three generative tasks: Knowledge-Based Question Answering (KB-QA), Dialogue, and Summarization.

A. Experimental Setup

For all three tasks, we prompt the ChatGPT to execute claim extraction, query generation, and multi-agent debate verification. The verification process is iterated a minimum of 2 rounds, and 10 snippets of evidence are extracted. The chosen transition method involved switching to the skeptic agent when the response was determined to be True.

1) Datasets and Baselines: The experimental datasets are derived from the following databases:

- Factool [12]: The Factprompts data comprises real-world questions with responses generated by ChatGPT, along with Factool-annotated claims extracted from these responses.
- HaluEval [16]: HaluEval constitutes a substantial collection of sampling-then-filtering generated and human-annotated hallucinated samples, serving as an evaluation metric for language model performance in recognizing hallucination.

We randomly selected 150, 50, and 150 samples from the three tasks of HaluEval for testing purposes. The selection of samples was contingent upon the complexity of task responses, with summarization outputs being more intricate. Owing to the necessity of decomposing summarization into a greater number of claims, the extracted quantity is comparatively smaller than that of the other two tasks. The positive and negative instances within the dataset were randomly sampled using a binary distribution with a probability of 0.5. The resulting data distribution is presented in Table I.

We compared the Factool method, the few-shot prompting method in HaluEval, the self-check method [12], and our approach.

2) Implementataion Details:

a) KB-QA: For complex QA data like Factool [12], we decomposed answers into atomic claims for multi-agent debate verification. If any claim has hallucinations, the original answer is deemed non-factual. As Factool lacks evidence, we used Google search to obtain evidences. Simpler QA data, as in HaluEval [16], with single-entity answers, were paired with questions for verification directly, utilizing dataset knowledge.

TABLE I
The Number of positive and negative samples in different datasets.

Datasets	Positive	Negative
Factool QA	23	27
HaluEval QA	75	75
HaluEval Summarization	25	25
HaluEval Dialogue	80	70

b) Summarization: The model-generated summary was treated as a response, decomposed into multiple claims, and each claim was verified individually. The corresponding document to the summary served as evidence. To mitigate excessively long input queries, each sentence of the document was encoded separately, along with the query. The top 10 most similar sentences were selected as evidence for the current claim.

c) Dialogue: Dialogue tasks posed challenges in claim extraction due to subjective viewpoints. To address this, a preprocessing step directed ChatGPT to remove subjective portions from its responses. Dialogue history and external knowledge supported claim extraction during verification.

B. Performance Analysis

The experimental results are presented in Table II and Table III. Table II shows the performance of our method on Factool [12], presenting results at both the claim and response levels. According to Table II, we can observe that our proposed method can consistently achieve optimal accuracy when compared to various approaches.

Table III displays the test results on the HaluEval [16] dataset, from which we can observe that: Our method demonstrates optimal accuracy, excelling in most metrics in all three tasks, Notably, in the three tasks of this dataset, our method exhibits a relatively low recall score. This can be attributed to our approach, which involves questioning claims verified as factual, thereby ensuring the precise detection of errors when claims are misclassified. However, this approach also results in misjudging some claims that inherently lack hallucinations as non-factual. This phenomenon is further elucidated in § IV-C0a.

C. Ablation Study

a) Transition Methods: We examined four transition methods' impact using 80 samples from the QA section of HaluEval [16]: transitioning to S_2 when the preceding state deems the claim non-hallucinated ($True \rightarrow$ **Skeptic**), transitioning to S_1 in the same case ($True \rightarrow$ **Trust**), and consistent transitions regardless of the preceding state's judgment. Results in Table IV show $True \rightarrow$ **Skeptic** outperformed across three metrics. This method challenges claims deemed factual, resulting in lower recall but higher precision, as discussed in § IV-B.

b) Minimum Rounds of Debate: We explored the influence of different numbers of minimum debate rounds on the outcomes. We examined three distinct tasks using the previously extracted HaluEval data [16], varying the number of minimum debate rounds from 0 to 3. Employing the $True \rightarrow$ **Skeptic** transition method, the results, illustrated in Figure 3, generally exhibit enhanced performance when the number of minimum rounds is set to 1 or 2, with a discernible decrease in efficacy when the number of minimum rounds is set to 3.

TABLE II

ACCURACY(%), RECALL(%), PRECISION(%), F1(%) OF FOUR METHODS ON DATASET FACTOOL [12] USED. CLAIM-LEVEL DENOTES THE RESULTS EVALUATED ON ALL ANNOTATED CLAIMS, AND RESPONSE-LEVEL DENOTES THE RESULTS EVALUATED ON THE ORIGIN RESPONSES. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**.

Method	Claim-Level				Response-Level			
	Acc.	R	P	F1	Acc.	R	P	F1
Self-Check (0)	75.54	**90.40**	80.00	**84.88**	54.00	**60.87**	50.00	54.90
Self-Check (3)	69.53	81.36	79.12	80.23	54.00	47.83	50.00	48.89
FACTOOL	74.25	73.45	**90.91**	81.25	64.00	43.48	66.67	52.63
Our Method	**77.68**	80.79	88.82	84.62	**72.00**	52.17	**80.00**	**63.15**

TABLE III

THE RESULTS FOR OUR METHOD AND BASELINE ON HALUEVAL [16] DATASET. WE CONDUCTED EXPERIMENTS ON THREE TASKS: QA, SUMMARIZATION, AND DIALOGUE. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**.

Method	QA				Summarization				Dialogue			
	Acc.	R	P	F1	Acc.	R	P	F1	Acc.	R	P	F1
HaluEval	56.00	77.33	54.21	63.74	58.00	**100.0**	54.35	**70.42**	68.00	**75.71**	63.10	68.83
FACTOOL	67.33	**86.67**	62.50	72.63	64.00	48.00	70.59	57.14	74.67	70.00	74,24	**72.06**
Ours	**70.67**	82.67	**66.67**	**73.81**	**70.00**	64.00	**72.73**	68.09	**76.00**	62.86	**81.48**	70.97

TABLE IV

COMPARISON OF DIFFERENT TRANSITION METHODS. WE EVALUATE THE INFLUENCE OF TRANSITION METHODS ON 80 QA SAMPLES, SETTING THE MINIMUM DEBATE ROUNDS TO 2. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**.

Method	Acc.	R	P	F1
Always **Skeptic**	65.00	84.21	59.26	69.57
Always **Trust**	68.75	84.21	62.75	71.91
True $\rightarrow$ **Trust**	67.50	**89.47**	60.71	72.34
True $\rightarrow$ **Skeptic**	**70.00**	86.84	**63.46**	**73.33**

TABLE V

COMPARISON WITH NON-GPT METHOD. WE COMPARE OUR METHOD WITH NON-GPT METHOD WECHECK ON THE FACTOOL DATASET. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**.

Method	Acc.	R	P	F1
Wecheck	65.23	64.41	86.36	73.78
Our method	**77.68**	**80.79**	**88.82**	**84.62**

c) Comparison with Non-GPT Method: In the multi-agent verification stage of the experiment in the Factool dataset, we employed the WeCheck [28] method to conduct an ablation study, showcasing the benefits of our approach. We held the initial two steps constant, utilizing the Factool method to extract claims and retrieve evidence. Employing the claim as the hypothesis and the evidence as the premise, instances with WeCheck scores greater than or equal to 0.5 were deemed factual. From the experimental results in Table V, we observed that compared to the non-GPT method, our approach exhibits significant advantages during the verification stage.

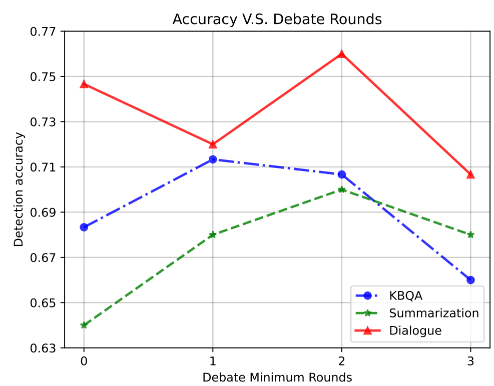

Fig. 3. **Comparison of Different Minimum Debate Rounds.** The **x**-axis represents different minimum debate rounds, whereas the **y**-axis signifies the corresponding detection accuracy.

V. CONCLUSION

In this paper, we introduce a versatile framework for hallucination detection and propose the Markov Chain-based multi-agent debate verification framework to improve the accuracy of hallucination detection in content generated by LLMs. Our proposed approach demonstrates its effectiveness through evaluations conducted on both the Knowledge Base Question Answering (KBQA) dataset and the randomly sampled HaluEval dataset. We posit that our method demonstrates a level of generalizability, enabling its adaptation to other post-processing hallucination detection or mitigation approaches for better performance.

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